

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                             |                              |
|-----------------------------|------------------------------|
| <b>Product Name</b>         | <u>Petroleum ether 40/60</u> |
| <b>CAS No</b>               | 64742-49-0                   |
| <b>Synonyms</b>             | Ligroine                     |
| <b>Recommended Use</b>      | Laboratory chemicals.        |
| <b>Uses advised against</b> | No Information available     |

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>L14649</b>                                                                               |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                  |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                    |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Flammable liquids

Category 2

#### Health hazards

Aspiration Toxicity  
Skin Corrosion/Irritation  
Specific target organ toxicity - (single exposure)

Category 1  
Category 2  
Category 3

#### Environmental hazards

Chronic aquatic toxicity

Category 2

### Label Elements

**Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor  
H304 - May be fatal if swallowed and enters airways  
H315 - Causes skin irritation  
H336 - May cause drowsiness or dizziness  
H411 - Toxic to aquatic life with long lasting effects

**Precautionary Statements****Prevention**

P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P273 - Avoid release to the environment

**Response**

P312 - Call a POISON CENTER or doctor if you feel unwell  
P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P331 - Do NOT induce vomiting  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse  
P391 - Collect spillage

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

This product does not contain any known or suspected endocrine disruptors

## Section 3 - Composition and Information on Ingredients

| Component                                               | CAS No     | Weight % |
|---------------------------------------------------------|------------|----------|
| Hydrocarbons, C6, isoalkanes < 5% n-hexane (Iso-Hexane) | 64742-49-0 | <100     |
| Hexane                                                  | 110-54-3   | -        |

## Section 4 - First Aid Measures

**Description of first aid measures**

|                                            |                                                                                                                                                                                                                                   |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                                                            |
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                                                        |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. Risk of serious damage to the lungs (by aspiration).                                                                 |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                                 |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.                               |
| <b>Self-Protection of the First Aider</b>  | Use personal protective equipment as required.                                                                                                                                                                                    |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                              |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                                                                                                   |

## Section 5 - Fire Fighting Measures

### **Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

### **Extinguishing media which must not be used for safety reasons**

Do not use a solid water stream as it may scatter and spread fire.

### **Specific Hazards Arising from the Chemical**

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### **Personal Precautions, Protective Equipment and Emergency Procedures**

#### **Emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

#### **Environmental Precautions**

Do not flush into surface water or sanitary sewer system.

#### **Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use

spark-proof tools and explosion-proof equipment.

**Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling****Advice on safe handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area.

**Incompatible Materials**

Strong oxidizing agents.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Control parameters****Exposure limits**

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

| Component                                               | New Zealand WEL                          | Australia                                | ACGIH TLV           | The United Kingdom                                                                      |
|---------------------------------------------------------|------------------------------------------|------------------------------------------|---------------------|-----------------------------------------------------------------------------------------|
| Hydrocarbons, C6, isoalkanes < 5% n-hexane (Iso-Hexane) |                                          |                                          |                     | RCP Isohexanes, TWA (8 h)<br>250 ppm, 1000 mg/m <sup>3</sup>                            |
| Hexane                                                  | TWA: 20 ppm<br>TWA: 72 mg/m <sup>3</sup> | TWA: 20 ppm<br>TWA: 72 mg/m <sup>3</sup> | TWA: 50 ppm<br>Skin | TWA: 72 mg/m <sup>3</sup><br>TWA: 20 ppm<br>STEL: 60 ppm<br>STEL: 216 mg/m <sup>3</sup> |

**Biological limit values**

**NZ** - Substances assigned Biological Exposure Indices in the New Zealand Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**ACGIH** - American Conference of Governmental Industrial Hygienists (ACGIH) TLVs® and BEIs®- Threshold Limit Values for Chemical Substances and Physical Agents & Biological Exposure Indices. 2022 Edition

| Component | New Zealand | Australia | ACGIH - Biological | United Kingdom |
|-----------|-------------|-----------|--------------------|----------------|
|-----------|-------------|-----------|--------------------|----------------|

|        |                                                  |  |                                                                                                          |  |
|--------|--------------------------------------------------|--|----------------------------------------------------------------------------------------------------------|--|
|        |                                                  |  | <b>Exposure Indices</b>                                                                                  |  |
| Hexane | 5 mg/L (urine) end of shift<br>(2,5-Hexanedione) |  | 0.5 mg/L<br>Medium: urine<br>Time: end of shift<br>Determinant:<br>2,5-Hexanedione without<br>hydrolysis |  |

**Appropriate engineering controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment**

**Eye Protection** Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection** Protective gloves

| Glove material             | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber, Viton (R). | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371 (or AUS/NZ equivalent)  
**Recommended half mask:-** Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
 When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system.

**Section 9 - Physical and Chemical Properties****Information on basic physical and chemical properties**

|                            |                           |
|----------------------------|---------------------------|
| <b>Physical State</b>      | Liquid                    |
| <b>Appearance</b>          | Colorless                 |
| <b>Odor</b>                | Petroleum distillates     |
| <b>Odor Threshold</b>      | No data available         |
| <b>pH</b>                  | No information available  |
| <b>Melting Point/Range</b> | No data available         |
| <b>Softening Point</b>     | No data available         |
| <b>Boiling Point/Range</b> | 40 - 70 °C / 104 - 158 °F |

|                                                |                                              |                                          |
|------------------------------------------------|----------------------------------------------|------------------------------------------|
| <b>Flammability (liquid)</b>                   | Highly flammable                             | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                               | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1 vol%<br><b>Upper</b> 7.4 vol% |                                          |
| <b>Flash Point</b>                             | -45 °C / -49 °F                              | <b>Method</b> - No information available |
| <b>Autoignition Temperature</b>                | 230 °C / 446 °F                              |                                          |
| <b>Decomposition Temperature</b>               | No data available                            |                                          |
| <b>Viscosity</b>                               | 0.45 mm <sup>2</sup> /s at 25 °C (ASTM D445) |                                          |
| <b>Water Solubility</b>                        | Insoluble                                    |                                          |
| <b>Solubility in other solvents</b>            | No information available                     |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                              |                                          |
| <b>Component</b>                               | <b>log Pow</b>                               |                                          |
| Hexane                                         | 4.11                                         |                                          |
| <b>Vapor Pressure</b>                          | 23 hPa @ 20 °C                               |                                          |
| <b>Density / Specific Gravity</b>              | 0.65                                         |                                          |
| <b>Bulk Density</b>                            | Not applicable                               | Liquid                                   |
| <b>Vapor Density</b>                           | >1 @ 101 kPa                                 | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | (liquid) Not applicable                      |                                          |

**Other information**

|                             |                                             |
|-----------------------------|---------------------------------------------|
| <b>Explosive Properties</b> | Vapors may form explosive mixtures with air |
| <b>Evaporation Rate</b>     | 13 (Butyl acetate = 1.0)                    |

## Section 10 - Stability and Reactivity

|                                         |                                                                                                       |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                                                            |
| <b>Stability</b>                        | Stable under normal conditions.                                                                       |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                              |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                              |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                         |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat, Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents.                                                                              |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).                                              |

## Section 11 - Toxicological Information

**Acute Effects****Information on likely routes of exposure****Product Information**

|                   |                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b> | Avoid breathing vapors or mists.                                                                                 |
| <b>Eyes</b>       | Avoid contact with eyes. May cause irritation.                                                                   |
| <b>Skin</b>       | Avoid contact with skin. May cause irritation. Prolonged skin contact may defat the skin and produce dermatitis. |
| <b>Ingestion</b>  | May be harmful if swallowed. Potential for aspiration if swallowed. Harmful if swallowed.                        |

**Numerical measures of toxicity**

**(a) acute toxicity;****Oral**

Based on available data, the classification criteria are not met

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

**Toxicology data for the components**

| Component                                               | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation              |
|---------------------------------------------------------|---------------------------|------------------------------|------------------------------|
| Hydrocarbons, C6, isoalkanes < 5% n-hexane (Iso-Hexane) | LD50 > 5000 mg/kg ( Rat ) | LD50 > 3160 mg/kg ( Rabbit ) | LC50 = 73680 ppm ( Rat ) 4 h |
| Hexane                                                  | LD50 = 25 g/kg ( Rat )    | LD50 = 3000 mg/kg ( Rabbit ) | LC50 = 48000 ppm ( Rat ) 4 h |

**(b) skin corrosion/irritation;**

Category 2

**(c) serious eye damage/irritation;**

Based on available data, the classification criteria are not met

**(d) respiratory or skin sensitization;****Respiratory**

Based on available data, the classification criteria are not met

**Skin**

Based on available data, the classification criteria are not met

**(e) germ cell mutagenicity;**

Based on available data, the classification criteria are not met

**(f) carcinogenicity;**

Based on available data, the classification criteria are not met

The table below indicates whether each agency has listed any ingredient as a carcinogen  
The classification as a carcinogen need not apply if the full refining history is known and it  
can be shown that the substance from which it is produced is not a carcinogen. This note  
applies only to certain complex oil derived substances in Annex I

| Component                                               | New Zealand | Australia | New South Wales | Western Australia | IARC | EU           | UK | Germany |
|---------------------------------------------------------|-------------|-----------|-----------------|-------------------|------|--------------|----|---------|
| Hydrocarbons, C6, isoalkanes < 5% n-hexane (Iso-Hexane) |             |           |                 |                   |      | Carc Cat. 1B |    |         |

**(g) reproductive toxicity;**  
**Reproductive Effects**

Based on available data, the classification criteria are not met  
California Proposition 65 Product is or contains a chemical which is a known or suspected reproductive hazard

**(h) STOT-single exposure;**

Category 3

**Results / Target organs**

Central nervous system (CNS)

**(i) STOT-repeated exposure;**

Based on available data, the classification criteria are not met

**Target Organs**

None known.

**(j) aspiration hazard;**

Category 1

**Symptoms / effects, both acute and delayed**

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## Section 12 - Ecological Information

**Ecotoxicity**

**Aquatic ecotoxicity**

The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component                                               | Freshwater Fish                                                  | Water Flea          | Freshwater Algae | Microtox |
|---------------------------------------------------------|------------------------------------------------------------------|---------------------|------------------|----------|
| Hydrocarbons, C6, isoalkanes < 5% n-hexane (Iso-Hexane) | LC50: = 8.41 mg/L, 96h semi-static, closed (Oncorhynchus mykiss) |                     |                  |          |
| Hexane                                                  | LC50: 2.1 - 2.98 mg/L, 96h flow-through (Pimephales promelas)    | EC50: 3.87 mg/L/48h |                  |          |

**Terrestrial ecotoxicity**

There is no data for this product

**Persistence and Degradability****Persistence**

Persistence is unlikely, based on information available.

**Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Hexane    | 4.11    | No data available             |

**Mobility**

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air

**Other adverse effects**

**Endocrine Disruptor Information**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste treatment methods****Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.

## Section 14 - Transport Information



**NZS 5433:2020**

**UN-No** UN1268  
**Proper Shipping Name** Petroleum distillates, n.o.s.  
**Hazard Class** 3  
**Packing Group** II

**IATA**

**UN-No** UN1268  
**Proper Shipping Name** Petroleum distillates, n.o.s.  
**Hazard Class** 3  
**Packing Group** II

**IMDG/IMO**

**UN-No** UN1268  
**Proper Shipping Name** Petroleum distillates, n.o.s.  
**Hazard Class** 3  
**Packing Group** II

| Component              | IMDG Marine Pollutant                                 |
|------------------------|-------------------------------------------------------|
| Hexane<br>110-54-3 (-) | IMDG regulated marine pollutant (Listed in the index) |

**Environmental hazards** Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code** Not applicable, packaged goods

**Special Precautions** No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

**Additional information** None known

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture****National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

| Component | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | IMDG Marine Pollutant                                 |
|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Hexane    |                                                                                           |                                                                                          | IMDG regulated marine pollutant (Listed in the index) |

#### Authorisation/Restrictions according to EU REACH

| Component                                               | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                         | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Hydrocarbons, C6, isoalkanes < 5% n-hexane (Iso-Hexane) | -                                                                   | Use restricted. See item 28. (see link for restriction details)<br>Use restricted. See item 29. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | -                                                                                                     |
| Hexane                                                  | -                                                                   | Use restricted. See item 75. (see link for restriction details)                                                                                                                                       | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

#### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                                               | CAS No     | NZIoC | AICS | EINECS    | ELINCS    | NLP | KECL     | IECSC | TCSI |
|---------------------------------------------------------|------------|-------|------|-----------|-----------|-----|----------|-------|------|
| Hydrocarbons, C6, isoalkanes < 5% n-hexane (Iso-Hexane) | 64742-49-0 | X     | X    | 265-151-9 | -         | -   | KE-25623 | X     | X    |
| Hexane                                                  | 110-54-3   | X     | X    | 203-777-6 | 438-390-3 | -   | KE-18626 | X     | X    |

| Component                                               | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|---------------------------------------------------------|------------|------|-----------------------------------------------|-----|------|-------|------|------|
| Hydrocarbons, C6, isoalkanes < 5% n-hexane (Iso-Hexane) | 64742-49-0 | X    | ACTIVE                                        | X   | -    | X     | -    | -    |
| Hexane                                                  | 110-54-3   | X    | ACTIVE                                        | X   | -    | X     | X    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

#### Legend

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**NZS 5433:2020** - Transport of Dangerous Goods on Land

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**PNEC** - Predicted No Effect Concentration

**OECD** - Organisation for Economic Co-operation and Development

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

#### Training Advice

Chemical incident response training.

#### Revision Date

02-May-2024

#### Revision Summary

New emergency telephone response service provider

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet